Supplementary Material for PREFLOW: Open Delegation of Long-Horizon Web Tasks for Blind and Low-Vision Users

Anonymous Author(s)

About This Supplement

The main paper argues that a Web agent can be trusted with open delegation only if it keeps, outside the language model's context, an explicit memory of what the user prefers and of what the visited pages show about each preference. This supplement follows that argument from the mechanism to the people it is meant to serve, in seven sections. Table S1 indexes the questions that a reader is likely to ask and where each is answered.

Numbers without the prefix S, such as Section 3, Table 2, or Appendix A.3, refer to the main paper. The analyses added here reuse the runs and the study reported in the main paper. The anonymous repository holds the released code and a video demonstration of PREFLOW.

Table S1: Questions a reader may ask, our short answers, and where this supplement supports them.

Question and short answer	Where
What does the explicit memory hold, and how does it change? Preference beliefs with their sources and, for every observed alternative, each belief's status with its evidence. A revision changes one belief and triggers a recheck of every recorded alternative.	S1; Table S2
Does PREFLOW see the benchmarks' goals or answers? No. No role reads WebShop's stored goals or rewards or EComAgentBench's scripted answers; settings are identical for both LLMs; held-out tasks give a similar accuracy.	S2.3, S4.4
Why WebShop and EComAgentBench? Both pose open delegation, execute the agent's actions, and score each preference separately. WebShop has published results for the agents we rerun; EComAgentBench tags each preference with its source.	S3
Where does the WebShop gain come from? Mostly from the explicit memory, which improves all four reward components; the *Commit Verifier Agent* adds option fixes.	S4.1
Does the gain survive a weaker LLM? Yes. From Gemini-3-Flash to DeepSeek-V3.2, LASER loses 13.5–16.0 points on every reward component; PREFLOW loses at most 3.6.	S4.2
Does the memory matter more in long tasks? Yes. In trajectories of more than 10 Web actions, it adds 9.0 points of complete satisfaction (7 instructions gained, none lost), versus 3.3 points otherwise.	S4.3
Are the differences beyond sampling noise? Yes for the paired comparison with LASER's policy on 500 instructions (+5.6 points, $p < 0.0001$); a 100-instruction sample alone is too small to tell.	S4.4
Is the EComAgentBench gain due to the LLM? No. With the same LLM, PREFLOW is correct in 25 of the 64 tasks in which the benchmark's own agent runs out of steps.	S4.5
Can a user who cannot see the page trust the report? Mostly. Before committing, the *Evidence Ledger* had flagged a doubtful preference in 83% of WebShop and 88% of EComAgentBench failures; silent failures occur in 6–8% of tasks.	S5; App. C
Does PREFLOW ask too much? At most four questions on EComAgentBench; all eight BLV participants rated the number of questions about right.	S4.5, S6.7
Did every BLV participant benefit? Trust rose for seven of eight and workload fell for six. The pattern holds without the participant with the largest differences and for the three who rarely or never use chat assistants.	S6.6
Does the design extend beyond shopping, and what remains open? The roles and the control flow stay; the task schema, evidence rules, and commit checks change. Threats to validity are listed with their mitigations.	S7

S1 Nora's Task, Step by Step

In brief. PREFLOW never relies on the language model to remember what Nora wants or what it has seen. Both live in an explicit memory that every later decision reads: what to ask, which desk to propose, and what to tell her before buying.

Figure S1 follows Nora's task (Section 1) through the six roles of Figure 1 and the steps of Algorithm 1. Table S2 shows the explicit memory after her revision, and Figure S2 shows the same task as Nora experiences it in the study interface (Section 5.3). The example is constructed, not recorded; Figure S3 in Section S3 follows a recorded run, and the video demonstration in the anonymous repository shows PREFLOW in use.

Three moments show why the memory must be explicit.

The question is grounded in observations (step 4). PREFLOW asks about price and assembly because the visited desks differ in both, and the instruction leaves both open (DG4). It does not ask about properties that no visited desk exhibits, and, in this task, it does not ask before it has observed desks.

The revision is local (step 7). Lowering the budget changes one belief, the price ceiling, and keeps the others (DG1). The *Evidence Ledger* then rechecks every recorded desk against the new ceiling (Algorithm 1, line 6) instead of relying on the model to recall which desk cost what (DG2). Desk 1, the only desk that supported every preference, now exceeds the ceiling, so the recommendation must change.

The preview names what is unmet (steps 8–9). After the revision, no desk supports every preference, and neither Desk 2 nor Desk 3 supports all that the other supports, the *Commit Verifier Agent* does not choose on its own; the trade-off goes to Nora (Section 3, Remarks). The preview is composed from the Evidence Ledger's statuses, so it states the one preference that the chosen desk contradicts, assembly (DG3, DG4). An agent that keeps preferences only in its context has no record against which to check such a preview.

S2 Implementation Details

In brief. The same six roles run in all three settings; what changes is how each role observes and acts. No role reads a benchmark's stored goals, rewards, or scripted answers.

This section complements Appendix A. Section S2.1 explains the task schemas, Table S4 shows how each role is implemented in the three settings, and Section S2.2 gives the decision rules; Table S5 lists the key settings, and the released code gives all other limits.

S2.1 Task Schemas

A Web page does not determine the task. The same page about Paris may serve a user who wants to learn the city's history and one who is planning a trip, and each needs something different from the agent. The *Task Router* therefore starts from what the

	Role	Step in Nora's task
1	*Task Router*	Nora asks: "I need a black folding desk." ROUTE returns transactional.obtain.
2	*Schema-Guided Planning Agent*	Preference beliefs: product *folding desk*; preference *color: black*; source: instruction.
3	*Executive Controller Agent*	Searches for folding desks and visits four: Desk 1 (black, fully assembled, $379.99), Desk 2 (black, requires assembly, $349.99), Desk 3 (white, fully assembled, $299.99), and Desk 4 (natural wood, requires assembly, $229.99).
4	*Dialogue Agent*	The visited desks differ in price and in whether they arrive assembled, and the instruction leaves both open, so the agent asks: "What's your budget, and would you like it fully assembled?" Nora: "Under $400. I'd prefer it fully assembled." UPDATE adds a price ceiling of $400 and *assembly: fully assembled*, both with source *reply*.
5	*Evidence Ledger*	Desk 1: supported {folding desk, black, price, fully assembled}. Desk 2: supported {folding desk, black, price}, contradicted {fully assembled}. Desk 3: supported {folding desk, fully assembled, price}, contradicted {black}. Desk 4: supported {folding desk, price}, contradicted {black, fully assembled}.
6	*Dialogue Agent*	Offers Desks 1–3 with what each meets and misses, and recommends Desk 1, which supports every preference. Nora asks what Desk 2 looks like and whether it is hard to assemble. The reply describes the photo ("The photo shows a matte black rectangular top on black metal legs…") and says what no observation addresses: "The listing doesn't say how hard assembly is."
7	*Schema-Guided Planning Agent, Evidence Ledger*	Nora: "My budget just changed. It's under $350 now." UPDATE revises the price ceiling, and the Evidence Ledger rechecks each desk's price: Desk 1 ($379.99) now contradicts it.
8	*Commit Verifier Agent*	No desk now supports every preference. Desks 2 and 3 each support one that the other does not, so the choice is a trade-off for Nora. The *Dialogue Agent* asks: "Which matters more to you: black, or fully assembled?" Nora: "Black. […] let's get the black office desk."
9	*Dialogue Agent*	Preview ρ_{pre}: "Folding Office Desk (Black), quantity 1, $349.99. It's black, folds, and is within your $350 budget. It doesn't meet your preference for fully assembled: it needs assembly. Shall I buy it for $349.99?" Nora approves.
10	Final action and report	The simulated purchase succeeds. Report ρ_F: "Done. I bought the Folding Office Desk (Black) for $349.99. It's a simulated purchase, so nothing was charged. Remember it needs assembly."

Figure S1: PREFLOW's roles in Nora's task (constructed example). Each row is one step of Algorithm 1 and names the role that performs it. Table S2 shows the explicit memory after step 7; Figure S2 shows the dialogue in the study interface.

Table S2: PREFLOW's explicit memory in Nora's task after her revision (step 7 in Figure S1; constructed example). Top: the preference beliefs $\widehat{P}$, each with its source. Bottom: the *Evidence Ledger* ℓ, with the status of each belief for each desk and the observation behind it.

Preference beliefs $\widehat{P}$

Belief	Value	Source	History
Product	folding desk	instruction	step 2
Color	black	instruction	step 2
Price	below $350	reply	below $400 at step 4; revised at step 7
Assembly	fully assembled	reply	step 4 ("I'd prefer")

Evidence Ledger ℓ (✓ supported, ✗ contradicted)

Desk	Product	Color	Price	Assembly
Desk 1	✓ title	✓ title	✗ $379.99	✓ "fully assembled"
Desk 2	✓ title	✓ title	✓ $349.99	✗ "requires assembly"
Desk 3	✓ title	✗ white	✓ $299.99	✓ "fully assembled"
Desk 4	✓ title	✗ natural wood	✓ $229.99	✗ "requires assembly"

After the revision, the Evidence Ledger rechecks all four prices against the new ceiling. Desk 1 now exceeds it, so no desk supports every preference, and the trade-off between color and assembly goes to Nora. Separately, the Evidence Ledger records what each desk's photo shows, such as Desk 2's black top and legs.

Table S3: Task schemas and their slots. Informational subtypes share the slots shown. Only transactional.obtain is evaluated.

Type	Subtype	Slots
Informational	Directed, undirected, find, list, advice	Topic; information need; other preferences; source or format
Navigational	To an informational site	Target site
Navigational	To a transactional site	Target service or site; intended action
Transactional	Obtain	Product or service; attributes; options; other preferences
Transactional	Download	File or software; version; format; cost or license
Transactional	Interact	Service or action; required inputs; preferences

user wants to do on the Web. Taxonomies of Web search intent supply the categories: Broder's informational, navigational, and transactional types [1] and Jansen et al.'s subtypes [11]. We found no taxonomy that specifies what the user needs from each category, so our task schemas add it: each schema lists the slots that the outcome of its tasks requires (Table S3). The history learner's task is informational, with a topic and an information need; the trip planner's is transactional, with the service, its attributes and options, and other preferences such as dates and budget. These slots are what the *Schema-Guided Planning Agent* fills and what the *Dialogue Agent*'s questions are about. A schema describes the intended outcome even when execution mixes types: a purchase also involves searching and navigating. Only transactional.obtain is evaluated (Appendix A.1); the other rows indicate how PREFLOW could extend to other tasks (Section S7).

S2.2 Decision Rules

This section adds to Appendix A the rules that decide each step; their limits and thresholds are in the released code.

Planning. In WebShop, the beliefs do not yet separate attributes from options; the *Evidence Ledger* classifies each preference by comparing it with product pages, and a price ceiling missing from the model's output is read from the instruction.

Executive Controller Agent. In WebShop, the controller follows LASER's policy [15]; invalid model outputs are repaired or retried. *Option handling* compares, before a purchase, the selected options with the preference beliefs for each option group, such as color or size. It selects a value only if the value matches a preference and appears on the page, and it replaces the controller's selection only if the new value keeps the preferences that the old one matched and matches more. *Recovery* starts from the controller's choice among the visited products, checks option conflicts, price, and product type against the beliefs, and may visit a few more products before buying the best one. In EComAgentBench, the controller retrieves the profile, runs the planned search queries (relaxing queries that return no results, and searching any store that the instruction names), ranks the results by how well their titles match the preferences, lets one model call shortlist products for inspection, and, after the questions, searches again and inspects a few more top-ranked products.

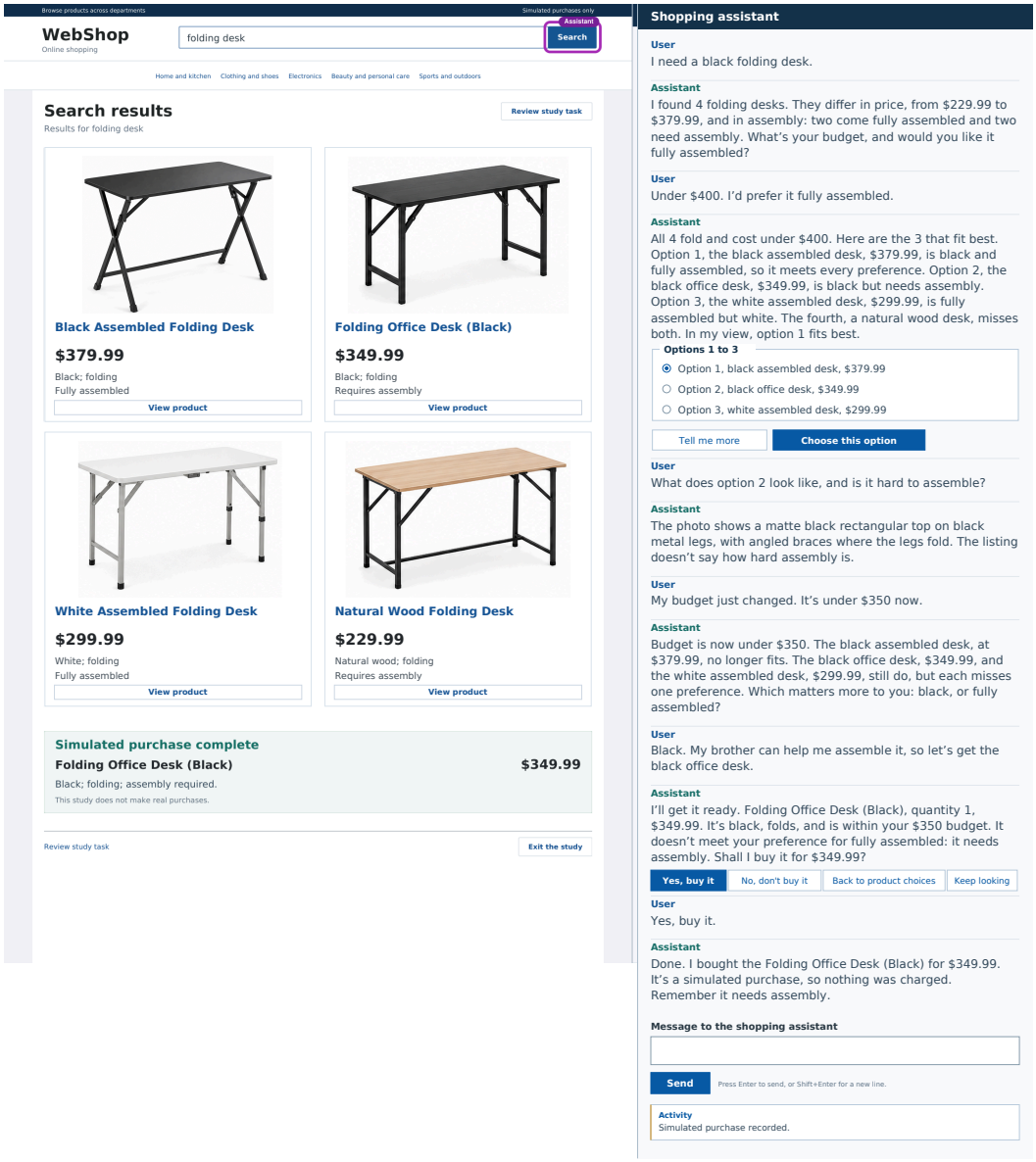


Figure S2: Nora's task in the study's Agent condition, with the shop beside PREFLOW's panel (scripted dialogue, not participant data). After finding four desks, PREFLOW asks about budget and assembly and recommends the desk that meets every preference. It describes another desk's photo and says what the listing does not state. When Nora lowers her budget, it rechecks the desks, presents the remaining trade-off, previews the purchase with the unmet preference, and buys only after explicit approval. Product images are generated illustrations; the purchase is simulated.

Evidence Ledger. In WebShop, $M_A(y)$ in the score of Appendix A.3 contains the preferences that y's title or text matches, and $M_O(y)$ those that exactly one of its options matches. On a results page, where options are not yet visible, the option terms are dropped, and the type estimate is reduced when no preference matches the title or options. The ranking redirects the controller to a clearly better-ranked result, surfaces first-page results that the controller skipped and products from further result pages, and issues reformulated searches when a preference that can be checked in titles matches no visited product. In EComAgentBench, the Evidence Ledger keeps every tool result and a snapshot of each inspected product (title, store, price, details, features, description, rating statistics, and review excerpts); the verification call gives each status an evidence quote and a reason, and deterministic checks then mark a violated numeric requirement as contradicted and a status whose quote does not appear in the evidence as unresolved.

Commit Verifier Agent. In WebShop, the four checks run in a fixed order, and the product changes at most once per task. (1) *Other*

Table S4: Implementation of each component in the three settings. The BLV study adds the interactive parts of Algorithm 1 (lines marked †), which neither benchmark supplies.

Component	WebShop	EComAgentBench	BLV study
Task Router	transactional.obtain (all tasks are shopping)	transactional.obtain	transactional.obtain
Schema-Guided Planning Agent	One model call; preference beliefs with product, preferences, and price ceiling	One model call; preferences with their source; profile and answers add preferences	As in WebShop; replies add, revise, or withdraw preferences
Executive Controller Agent	LASER's policy over Web pages	Fixed stages over the benchmark's tools	As in WebShop; also searches further when the user asks for more options
Evidence Ledger	Matches between preferences and product pages; ranking and further search	Statuses with evidence quotes from the verification call, checked deterministically	As in WebShop; also records, separately, what product photos show
Commit Verifier Agent	Four deterministic checks before purchase	Model call assesses each preference; challenger check	Never buys on its own: purchase only after the user's approval, and never a product the request rules out
Dialogue Agent	Off (no user)	Up to four questions about the schema's slots, without a model call	Questions (some, such as size or budget, before the first search), trade-offs, photo descriptions, preview and approval, report
Final action	Purchase	Recommendation	Simulated purchase after approval, or none

Table S5: Key settings; WebShop's are identical for both LLMs. All other limits and thresholds are in the released code.

Setting	Value
WebShop	
LLMs	Gemini-3-Flash; DeepSeek-V3.2
Temperature; top-p	0; 1
Output tokens: planning; controller	400; 200
Controller main-loop iterations	13
EComAgentBench	
LLM; temperature	Gemini-3-Flash; 1.0
Evaluator; temperature	Gemini-3.1-Pro; 1
Model calls per task	Up to 4
Questions per task	Up to 4
Products inspected per task	Up to 16
BLV study	
LLM (text and photos)	Gemini-3.7-Flash
Temperature	Provider default (not settable)

search results: if the proposed product does not match every preference or lacks a required option, the Commit Verifier Agent visits a few other products on the current results page that are within the price ceiling and have at least the same type estimate, and switches to the one whose page matches the most preferences, provided that it matches more than the proposed product and loses none of its matches; failing that, it may switch to one whose full-page text, including WebShop's repeated instruction, matches more preferences, under the same conditions and a score margin. (2) *Visited products*: it switches to a visited product that matches every preference that the proposed product matches, and at least one more. (3) *Best-ranked product*: it switches to the Evidence Ledger's top-ranked product if this product is within the price ceiling and either has a higher type estimate and a clearly higher score, or the same type estimate and more options that match preferences. (4) *Missing option*: if the proposed product lacks a required option, it switches to a recorded product that offers the option, has no conflicting option, is within the price ceiling, and has at least the same type estimate and a similar score. Checks 1 and 2 follow the rule of Section 4.4; the fallback of check 1 and checks 3 and 4 may lose a match elsewhere, within their conditions. In EComAgentBench, the challenger check runs only if a preference is not supported: it searches again for the unsupported preferences and inspects a few more top-ranked products, and a second model call compares the selection with all inspected products; its choice is kept unless it introduces a numeric contradiction that the selection did not have.

Dialogue Agent. In EComAgentBench, after the shortlisted products are inspected, Ask asks up to four short questions about the slots of transactional.obtain: usually first about the budget ("What is your budget for this item?"), then about attributes and other preferences, such as special features, ratings, or connectivity ("Do you need a specific connectivity technology?"). Figure S3 shows four.

S2.3 What PREFLOW Does Not See

An agent could gain simply by reading what the evaluator checks; PREFLOW cannot.

No stored goals or rewards. In WebShop, the *Schema-Guided Planning Agent* sees only the instruction, and the *Evidence Ledger* and the *Commit Verifier Agent* see only the preference beliefs and the pages. None of them reads WebShop's stored attributes, options, price ceiling, or reward (Appendix A.3). The Evidence Ledger's ranking score has the same structure as WebShop's reward because both express the notion of success in Section 3: the share of preferences that an outcome satisfies, gated by the product type. The score is computed from beliefs and observations only.

No echo of the instruction. WebShop prints the instruction at the top of every page. The Evidence Ledger removes this header before matching, so a product cannot appear to satisfy a preference merely because the instruction is shown on its page. Only the fallback for the Commit Verifier Agent's first check uses the full page (Section S2.2); the header contains the instruction that every agent receives, not the stored goal.

No scripted answers. In EComAgentBench, PREFLOW obtains clarification preferences only by asking; it never reads the simulated user's script or the target product (Appendix A.5).

No per-LLM tuning. All limits and thresholds (Table S5 and the released code) are identical for Gemini-3-Flash and DeepSeek-V3.2, yet the gain over LASER is larger with DeepSeek-V3.2 (Section S4.2).

A held-out split. PREFLOW was developed on 250 EComAgentBench tasks. On 100 tasks from the 412-task held-out split, never used for development, its accuracy is 45.0% (95% interval [35.6, 54.8]), consistent with its 48.3% on all 662 tasks ([44.6, 52.1]; Section S4.4).

S3 Choice of Benchmarks and a Recorded Run

In brief. RQ1 needs executable shopping tasks under open delegation, with an evaluator that checks each preference separately. WebShop provides this with fully stated preferences and published results for the agents we compare against; EComAgentBench provides it with partly stated preferences, each tagged with its source, and results for seven current agents.

S3.1 Criteria and Candidates

RQ1 requires open delegation, an executable Web environment or tools, automatic evaluation of the final outcome, long-horizon tasks stated in natural language, separate checks of each requested property, and fully or partly stated preferences (Section 3). Table S6 compares the benchmarks we considered against these criteria. *Open delegation*: the user states the goal, and the agent chooses the actions [3]. RealWebAssist poses closed delegation instead: users give one instruction at a time, and each predicted action is scored against the recorded one [24]. *Dynamic interaction*: the agent's actions are executed and return new observations; recorded traces do not qualify. *Outcome evaluation*: the final answer or state is checked automatically against the task's goal, possibly by a model. *Multi-step*: tasks require several interdependent actions. *Natural-language tasks*: goals are stated in natural language, not as structured fields. *Property checks*: each requested property is checked separately, even if the checks are combined into one score.

Two considerations decided the choice. First, WebShop is the standard benchmark for shopping tasks with fully stated preferences, and it is the benchmark on which the five agents we rerun (Act, ReAct, ASH, AskAct, LASER) report their results, which allows both a published and a same-LLM comparison. It is not saturated: its human shoppers satisfy all preferences in 50.0% of test instructions (Table 3). Second, among the benchmarks with partly stated preferences that meet all other criteria, EComAgentBench tags each preference with its source—instruction, profile, or clarification answer—and reports results for seven current agents [6]. Source tags are what RQ1b measures: they separate the preferences that the agent must obtain from those that the instruction states. Evaluating PREFLOW on the other candidates in Table S6 is future work.

S3.2 A Recorded EComAgentBench Run

Figure S3 follows PREFLOW through one task, chosen because it shows every stage. An answer to a question adds a preference that the first selection contradicts; the challenger check then finds a product that supports all nine preferences, and it is the target.

S4 Where the Gains Come From, and How Far They Hold

In brief. The explicit memory, not the LLM, carries most of the gain. It improves every reward component, matters most in long trajectories, and keeps the share of satisfied preferences nearly unchanged under a weaker LLM. On EComAgentBench, PREFLOW's advantage over the benchmark's own agent comes from always reaching a recommendation that its verification supports.

S4.1 Which Preferences Does PREFLOW Satisfy More Often?

Figure 2 of the main paper decomposes WebShop's reward into its four components (Appendix B.1) for each agent and LLM on the 100 sampled instructions; Figure S4 does the same for the ablation on all 500.

With Gemini-3-Flash, PREFLOW exceeds LASER on product type (93.4% versus 91.8%) and options (67.7% versus 62.7%), ties it on price (99.0%), and is lower on attributes (80.7% versus 83.0%). On all 500 instructions, by contrast, the explicit memory raises attribute satisfaction over LASER's policy, from 76.4% to 80.0% (Figure S4). With DeepSeek-V3.2, PREFLOW is higher than LASER on all four components: options (64.1% versus 48.7%), price (98.0% versus 83.0%), product type (89.8% versus 76.5%), and attributes (82.3% versus 69.5%). For every agent and both LLMs, options have the lowest satisfaction.

In the ablation, the explicit memory (V1) improves all four components over V0: type by 1.4 points, price by 1.0, options by 2.3, and attributes by 3.6. The *Commit Verifier Agent* (V2) improves only options, from 66.2% to 69.3%. Its role is narrow by design: it catches products whose required option, such as a size or color, is visible only on the product page.

S4.2 Does the Gain Survive a Weaker LLM?

Table S7 shows how much each agent loses on each reward component when the LLM changes from Gemini-3-Flash to DeepSeek-V3.2 on the same 100 instructions. Every other agent loses at least 5.7 points on every component, LASER between 13.5 and 16.0. PREFLOW loses at most 3.6, and its attribute satisfaction rises by 1.6.

The price component shows the mechanism most clearly. A price ceiling read in the instruction must still hold at the moment of purchase, often many pages later. With DeepSeek-V3.2, LASER exceeds the ceiling in 17% of instructions; PREFLOW, which stores the ceiling as a belief and checks it before purchase, exceeds it in 2%.

In complete satisfaction, PREFLOW, LASER, and ASH each lose 6 of the 100 instructions (Table 2). The difference lies in how they fail: PREFLOW's mean satisfaction drops by only 3.8 points, LASER's by 13.5, so the instructions that PREFLOW no longer satisfies completely remain near misses.

S4.3 Does the Memory Matter More in Long Tasks?

Section 6.1 reports that the explicit memory helps most when V0 takes more than 10 Web actions, twice the median. Table S8 adds the complementary subset. Long trajectories are the hard ones: V0 satisfies all preferences in 28.2% of them, versus 54.3% of the others. There, the explicit memory adds 9.0 points without losing a single instruction; elsewhere, it adds 3.3 points. The gain is significant in both subsets, but nearly three times as large where a context-only agent must carry the preferences across many pages, which is the situation the explicit memory is designed for.

The number of preferences tells a related story. Complete satisfaction is lower with four or more preferences (46.1–61.1%) than with three (68.6%; Table S9), but mean satisfaction stays between 73.1% and 81.8% at every size: PREFLOW rarely fails entirely, even

Table S6: Benchmarks considered for RQ1, as described in the cited publications through September 2026. A cross means that the criterion is not part of the published protocol, not that the environment cannot be extended. Bold rows are the benchmarks used in this paper.

Benchmark	Environment	Open delegation	Dynamic interaction	Outcome evaluation	Multi-step	Natural-language tasks	Property checks	Fully stated preferences	Partly stated preferences
FormWoB/QAWoB [20]	Cached Web	✓	✓	✓	✓	✗	✓	✓	✗
VirtualHome [17]	Household simulator	✓	✓	✗	✓	✓	✗	✓	✗
AndroidEnv [21]	Mobile apps	✓	✓	✓	✓	✗	✗	✗	✗
Mind2Web [5]	Recorded Web traces	✓	✗	✗	✓	✓	✗	✓	✗
WebArena [25]	Hosted Web	✓	✓	✓	✓	✓	✓	✓	✗
WebLINX [14]	Recorded Web dialogue	✓	✗	✗	✓	✓	✗	✗	✓
RealWebAssist [24]	Recorded Web dialogue	✗	✗	✗	✓	✓	✗	✗	✓
REAL [8]	Hosted Web	✓	✓	✓	✓	✓	✓	✓	✗
Mind2Web 2 [9]	Live Web research	✓	✓	✓	✓	✓	✓	✓	✗
AgenticShop [12]	Live Web	✓	✓	✓	✓	✓	✓	✓	✗
WebMall [16]	Hosted shops	✓	✓	✓	✓	✓	✗	✓	✗
ChatShop [4]	Search tools, dialogue	✓	✓	✓	✓	✓	✓	✓	✓
PROCLARE [23]	Retrieval, dialogue	✓	✓	✓	✓	✓	✗	✗	✓
UserBench [18]	Travel tools, dialogue	✓	✓	✓	✓	✓	✗	✗	✓
Persona2Web [13]	Live Web, history	✓	✓	✓	✓	✓	✓	✓	✓
ACWorld [7]	Commerce tools	✓	✓	✓	✓	✓	✓	✓	✓
WebShop [22]	Simulated shop	✓	✓	✓	✓	✓	✓	✓	✗
EComAgentBench [6]	Shopping tools, dialogue	✓	✓	✓	✓	✓	✓	✗	✓

	Stage	What happens
1	Task	Instruction: "I need a surveillance camera kit that comes with the 8 Bullet Cameras option and shoots in 1080p resolution. Ensure it includes 3 TB of storage, and only show me systems with at least 3 stars and at least 20 ratings." The profile requires a corded electric power source and compatibility with "Cameras LNAVA". Two further preferences are revealed only if asked.
2	*Schema-Guided Planning Agent*	Seven preferences: from the instruction, *8 bullet cameras*, *1080p*, *3 TB storage*, *at least 3 stars*, and *at least 20 ratings*; from the profile, *corded electric* and *compatible with Cameras LNAVA*.
3	*Executive Controller Agent*	Searches for camera kits and inspects the details, ratings, and reviews of seven.
4	*Dialogue Agent*	Asks four questions. "What is your budget for this item?" — "Hmm, I'm not sure about that." "Are there any special features you require?" — "I definitely need it to have night vision so I can keep an eye on things after dark." "Do you need a specific connectivity technology?" — "I want a wired system because I feel like it's a lot more reliable than relying on Wi-Fi." "Do you require a specific color?" — "Hmm, I'm not sure about that." Two answers add the preferences *night vision* and *wired*; a second search adds three more kits.
5	*Commit Verifier Agent*: selection	Selects the ZOSI 8CH 1080P Wireless Security Camera System. Eight preferences are supported, including night vision ("80ft Night Vision"); *wired* is contradicted ("Connectivity Technology: Wireless").
6	*Commit Verifier Agent*: challenger check	Searches again for a wired system and inspects six more kits. The final model call compares all inspected kits and chooses the LaView 8 1080P IP Camera Security System, which supports all nine preferences ("Wired"; "3 TB"; "Cameras LNAVA"; "see up to 100 feet at night"). It introduces no numeric contradiction, so it replaces the ZOSI kit.
7	Recommendation and evaluation	PREFLOW recommends the LaView kit, which is the target product, after 4 questions, 16 inspected products, and 4 model calls.

Figure S3: PREFLOW on one EComAgentBench task with Gemini-3-Flash (recorded run of the reported version). Quotes are from the run's records; the user's answers come from the benchmark's script.

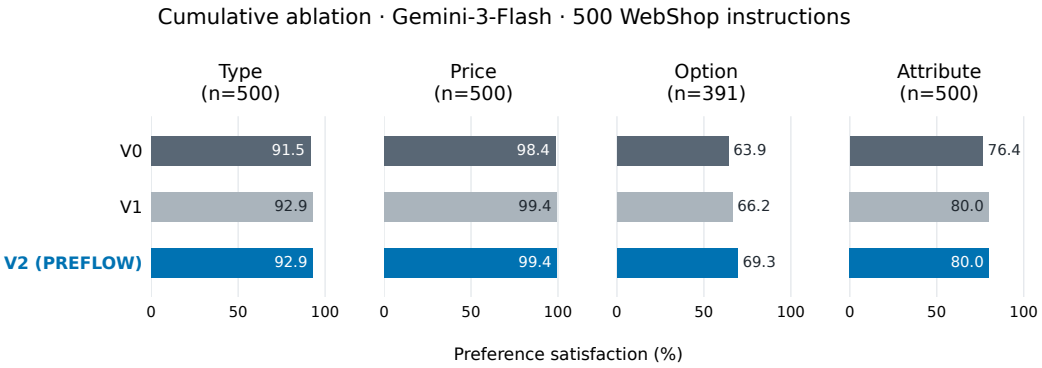


Figure S4: Satisfaction of each reward component in the cumulative ablation on the 500-instruction test split with Gemini-3-Flash. V0 is LASER's policy, V1 adds the explicit memory (preference beliefs and *Evidence Ledger*), and V2 adds the *Commit Verifier Agent*. Option satisfaction is averaged over the 391 instructions that require an option.

Table S7: Change from Gemini-3-Flash to DeepSeek-V3.2 on the same 100 WebShop instructions, in percentage points (DeepSeek-V3.2 minus Gemini-3-Flash). Components from Figure 2; Mean and Complete from Table 2.

Agent	Type	Price	Option	Attribute	Mean	Complete
Act	−42.4	−47.0	−27.5	−39.0	−33.9	−16
ReAct	−36.0	−37.0	−22.4	−33.5	−30.3	−20
ASH	−11.1	−12.0	−5.7	−11.5	−9.1	−6
AskAct	−49.6	−52.0	−28.8	−47.0	−41.4	−27
LASER	−15.3	−16.0	−14.0	−13.5	−13.5	−6
PREFLOW	**−3.6**	**−1.0**	**−3.6**	**+1.6**	**−3.8**	−6

Table S8: Ablation V0 → V1 on the 500 WebShop instructions, split by the number of Web actions that V0 takes (median 5). Complete satisfaction in %; exact McNemar tests within each subset (Appendix B.2). Counts for the shorter subset follow from Table 8 and the longer subset.

V0's Web actions	N	V0	V1	Gained	Lost	p
At most 10	422	54.3	57.6	19	5	0.007
More than 10	78	28.2	37.2	7	0	0.016
All	500	50.2	54.4	26	5	0.0002

Table S9: PREFLOW's preference satisfaction by the number of preferences in the instruction, on all 500 WebShop test instructions (Gemini-3-Flash). Table S10 breaks it down by component.

Preferences	N	Complete (%)	Mean (%)
3	70	68.6	79.4
4	208	61.1	81.8
5	167	46.1	73.1
6 or more	55	49.1	77.5

Table S10: PREFLOW's satisfaction of each reward component by the number of preferences in the instruction, on the 470 WebShop test instructions whose target product obtains full reward (Gemini-3-Flash). All values in %.

Preferences	N	Complete	Mean	Type	Attribute	Option	Price
3	70	70.0	80.6	91.1	72.9	—	98.6
4	196	58.2	80.8	94.7	85.7	67.5 (166)	99.5
5	155	45.2	73.8	93.2	76.6	64.5 (148)	98.1
6 or more	49	53.1	82.0	100.0	87.4	70.5 (48)	100.0

with six or more preferences. Table S10 breaks this down by reward component on the 470 instructions whose target product obtains full reward. Option satisfaction is averaged only over instructions that require at least one option; their number is in parentheses.

S4.4 How Large Is the Uncertainty?

WebShop runs use temperature 0 and EComAgentBench runs temperature 1.0 (Table S5); all intervals reflect variation across instructions or tasks, not across repeated runs (Appendix B.2). Table S11 gives 95% intervals for the main estimates, and two conclusions follow.

Table S11: Main estimates with 95% intervals. Single proportions: Wilson intervals, computed from the reported percentages for published results. Paired differences: bootstrap intervals and exact McNemar tests (Appendix B.2).

Estimate	Value (%)	95% interval
WebShop, complete satisfaction		
PREFLOW, 100 instructions, Gemini-3-Flash	55.0	[45.2, 64.4]
LASER, 100 instructions, Gemini-3-Flash	50.0	[40.4, 59.6]
PREFLOW, 100 instructions, DeepSeek-V3.2	49.0	[39.4, 58.7]
LASER, 100 instructions, DeepSeek-V3.2	44.0	[34.7, 53.8]
PREFLOW, 500 instructions	55.8	[51.4, 60.1]
V0 (LASER's policy) → PREFLOW, 500, paired	+5.6	[+3.0, +8.2]
EComAgentBench, accuracy		
PREFLOW, 662 tasks	48.3	[44.6, 52.1]
GPT-5.4, 662 tasks	47.0	[43.2, 50.8]
Kimi K2.6, 662 tasks	46.4	[42.6, 50.2]
Claude Opus-4.6, 662 tasks	57.1	[53.3, 60.8]
PREFLOW, 100 held-out tasks	45.0	[35.6, 54.8]
Benchmark agent, 100 held-out tasks, same LLM	23.0	[15.8, 32.2]

Table S12: Correct recommendations on the 100 held-out EComAgentBench tasks with the same LLM (Gemini-3-Flash), split by whether the benchmark's agent finishes within its 100 steps. Derived from Section 6.2.

		Correct	
Tasks in which the benchmark's agent	N	Benchmark agent	PREFLOW
finishes	36	23	20
runs out of steps	64	0	25
All tasks	100	23	45

First, on 100 instructions a single proportion is uncertain by about ±10 points, so PREFLOW's 5-point lead over LASER on the 100 sampled instructions is not conclusive by itself. The paired comparison on all 500 instructions, in which V0 runs LASER's policy with the same LLM, is conclusive: +5.6 points [+3.0, +8.2], with 36 instructions gained and 8 lost ($p < 0.0001$; Table 8).

Second, on EComAgentBench, PREFLOW's accuracy cannot be distinguished from that of GPT-5.4 or Kimi K2.6 and is below that of Claude Opus-4.6. What distinguishes PREFLOW is that it reaches this accuracy with 3.7 model calls per task instead of 25.1–38.2 for these three agents (Table 5).

S4.5 Where Does the EComAgentBench Gain Come From?

With the same LLM on the same 100 held-out tasks, PREFLOW is correct in 45 tasks and the benchmark's own agent in 23 (Table 5; McNemar $p < 0.001$). Table S12 shows where the difference arises. The benchmark's agent exhausts its 100 steps in 64 tasks—in 18 of them after it had already retrieved the target product—and is correct in none of them. PREFLOW is correct in 25 of these 64. On the 36 tasks that the benchmark's agent completes, it is correct slightly more often (23 versus 20). PREFLOW's gain is therefore one of reliability: its fixed stages always end with a recommendation that its verification supports, with 21 times fewer model calls.

How broadly does PREFLOW search? Table S13 compares the search effort of both agents on the same 100 tasks. The benchmark's agent rarely looks beyond the first page of results and spends most of its steps on review searches; PREFLOW looks beyond the first

Table S13: Search effort on the 100 held-out EComAgentBench tasks with the same LLM (Gemini-3-Flash).

	PREFLOW	Benchmark agent
Tasks with a result page beyond the first	96	4
Products inspected (median)	15	5
Review searches (median)	11	54.5
Tool calls (mean)	49.8	75.2
Model calls (mean)	3.6	75.2
Tasks ending at the step limit	0	64

Table S14: EComAgentBench accuracy (%) on all 662 tasks by the number of requirements that the instruction does not state. The average over the seven published agents and GPT-5.4's values are from [6]. No single difference between PREFLOW and GPT-5.4 is statistically significant.

Unstated requirements	Tasks	Published average	GPT-5.4	PREFLOW
0–1	215	51.8	59.1	59.1
2	335	36.1	44.5	46.3
3 or more	112	27.2	31.2	33.9

page in almost every task and inspects three times as many products with fewer tool calls. Looking further matters: on all 662 tasks, 20 of PREFLOW's 320 correct recommendations ranked below the first results page in every search in which they appeared, so an agent that reads only first pages would not have seen them. Products found that deep are harder to judge: 20 of the 94 such recommendations were correct, against 258 of the 515 found on a first page.

Asking after observing also pays off. PREFLOW asks 4.0 questions per task, the most of any agent and its maximum, and satisfies the largest share of clarification preferences (72.2%; Table 5). With the same LLM, the benchmark's agent asks 0.2 questions per task and satisfies 27.0% of clarification preferences, against 66.5% for PREFLOW. Each task holds two clarification preferences, which the simulated user reveals only when asked, so asking is how PREFLOW obtains them.

Table S14 breaks down accuracy on all 662 tasks by the number of requirements that the instruction does not state, using the groups of the benchmark's own analysis [6]. PREFLOW ties GPT-5.4 when at most one requirement is unstated and leads by 1.8 and 2.7 points with two and with three or more.

S5 Can a User Trust What PREFLOW Reports?

In brief. Before committing, the *Evidence Ledger* knows which preferences it could not confirm. It had flagged a contradicted or unresolved preference in 83% of WebShop failures and 88% of EComAgentBench failures, so a report would have warned the user; only 7.6% and 6.2% of tasks ended in a silent failure.

For a user who cannot see the page, what the agent says about a purchase matters as much as the purchase. PREFLOW's preview and report state the Evidence Ledger's statuses at commitment (Section 4.5), so two questions decide their value: when the Evidence Ledger supports every preference, is the outcome right; and when the outcome is wrong, had the Evidence Ledger flagged it?

Neither benchmark shows the report to a user, but the Evidence Ledger's state at commitment is recorded. Figure 3 of the main paper crosses this state with each benchmark's evaluation: (a) all 500 WebShop test instructions (full PREFLOW, Gemini-3-Flash; 55.8% complete, as in Table 4) and (b) all 662 EComAgentBench tasks (48.3% correct, as in Table 5), where a model call assigns the statuses (Appendix A.3).

Full support is informative. When the Evidence Ledger supports every preference belief, 107 of 145 WebShop purchases (73.8%) satisfy all stored preferences, against 172 of the other 355 (48.5%). On EComAgentBench, 217 of 258 such recommendations (84.1%) are correct, against 103 of the other 404 (25.5%).

Most failures are flagged before committing. The Evidence Ledger had recorded a contradicted or unresolved preference in 183 of the 221 WebShop purchases that miss a stored preference (82.8%) and in 301 of the 342 incorrect EComAgentBench recommendations (88.0%). A report would have told the user which preference was in doubt.

Silent failures are rare. Only 38 WebShop tasks (7.6%) and 41 EComAgentBench tasks (6.2%) end in a failure although the Evidence Ledger supported every belief.

String matching errs on the side of caution. Of the 279 complete WebShop purchases, 172 (61.6%) were not fully supported at commitment. The WebShop Evidence Ledger matches preferences against page text (Appendix A.3), so a preference remains unresolved when the page does not state it in matching words, even if the product has it. The model-based Evidence Ledger in EComAgentBench is less cautious: 103 of the 320 correct recommendations (32.2%) were not fully supported.

Silent failures point back to the formalization of Section 3. The Evidence Ledger checks observations against the beliefs $\widehat{P}$, not against the user's preferences P, so it cannot flag a preference that the beliefs omit or misread. A report can be no more accurate than the beliefs it is checked against, which is why PREFLOW records each belief with its source and lets the user correct it (DG1).

An agent that keeps preferences only in its context, such as LASER, commits without a per-preference record and has nothing against which a report could be checked. For BLV users, who cannot inspect the product page themselves, this is the difference between a report that says what is in doubt and a confirmation that only says what was bought.

S6 The BLV Study in Detail

In brief. Eight BLV participants each shopped once on their own and once with PREFLOW. With PREFLOW, trust rose for seven and workload fell for six, and seven preferred it. The gains were not confined to experienced chat-assistant users, and all eight rated the number of questions about right.

This section gives the materials and full results of the study in Section 5.3; Appendix B.4 reports the ethics approval and consent.

S6.1 Participants and Procedure

Participants were recruited through organizations of blind and low-vision people (names omitted for anonymous review). Eight adults who are blind or have low vision and use a screen reader took part (Table S15). They received compensation by bank transfer as a thank-you for their time. They were told that the study evaluates two ways of shopping, not them; that there are no right or wrong answers; and that honest criticism is welcome. Interaction logs and

Table S15: Participants. Proficiency with their screen reader is self-rated (1–5); chat assistants: how often they use conversational AI assistants.

	Age	Vision (duration)	Screen reader	Prof.	Chat assistants
P1	25–34	Blind, light perception (since birth)	NVDA	5	Daily
P2	25–34	Blind, light perception (since birth)	NVDA	5	Daily
P3	45–54	Blind (over 20 years)	JAWS	5	Daily
P4	25–34	Blind (since birth)	NVDA	5	Daily
P5	45–54	Blind, light perception, peripheral loss (11–20 years)	JAWS, braille display	4	Tried once
P6	35–44	Low vision, central loss (6–10 years)	VoiceOver, magnifier	3	Occasionally
P7	55–64	Blind (over 20 years)	JAWS	4	Never
P8	18–24	Blind, light perception (since birth)	VoiceOver, dictation	5	Daily

Table S16: Steps of a study session.

	Step	Content
1	Consent and intake	Informed consent; intake questionnaire
2	Orientation	Introduction to the shop with the participant's own assistive technology
3	First task	Manual or Agent condition; a short practice with PREFLOW precedes the Agent task
4	Questionnaire	SEQ, NASA-TLX, control, SUS, trust, product satisfaction; final preferences
5	Break	
6	Second task	The other condition and the other task
7	Questionnaire	As in step 4
8	End of session	Message items, preferred condition, open questions

questionnaire responses are stored under participant codes. An intake questionnaire records age range, vision status and its duration, assistive technology and experience with it, browser, online shopping frequency, and prior use of voice and chat assistants.

Table S16 lists the steps of a session. Condition order (Manual first or Agent first) and the assignment of the two tasks to the conditions are counterbalanced, giving four sequences. Sessions were held online on a video-call platform of the participant's choice, one session at a time, with the moderator present throughout. A short practice with PREFLOW, which is not measured, precedes the Agent task. A session took about 60 minutes.

S6.2 Interfaces

Both conditions use the same shop, built on WebShop's product catalog and search engine [22] (Figure S5). A search returns up to 50 products in pages of 10. Each product has a fixed purchase price that is the same in both conditions, and purchases are simulated. In the Manual condition, participants search, open product pages, select options, and buy on their own. The Agent condition adds PREFLOW's conversation panel next to the shop, and participants can still browse the shop directly. In this panel, the *Dialogue Agent* asks for missing preferences, presents trade-offs, requests approval, and reports the outcome. Table S17 lists how its interface serves screen-reader and low-vision users. The preview ρ_{pre} states the product, selected options, exact price, and supported, contradicted, and unresolved preferences. Approval applies only to this product, these options, and this price; any later change cancels it. Figure S2 shows a scripted conversation in this interface.

Table S17: How the Agent condition's interface serves screen-reader and low-vision users (target: WCAG 2.2 AA).

Need	What the interface does
Hear replies without losing one's place	Each reply is announced once without moving keyboard focus, with an optional sound; each message is a heading, so the latest reply is one keystroke away.
Know what PREFLOW is doing	A status region gives a progress line at least every 5 s and reports problems; an activity list records searches, products opened, and options selected, never the model's reasoning.
Stay in charge of the shop	The panel is a landmark placed before the shop. Pages that PREFLOW opens replace the shop area without reloading. Participants can browse or select options themselves, and PREFLOW continues from that state; replies link the products they mention.
Answer quickly	A question offers up to seven reply buttons, each sending the same text that the participant could type.
Compare products	Comparisons are spoken and also given as a data table with row and column headers.
Know what products look like	PREFLOW describes product photos on request and marks what comes from a photo (Appendix A.3).
Low vision	The layout reflows to one column; text contrast is at least 7:1; focus is clearly visible; forced colors and reduced motion are respected.

Both conditions load the same shop pages, so every accessibility fix applies to both, such as a labeled search box, results named by product title instead of product code, ratings given in text, and no meaning conveyed by color alone. The shop's ordinary friction is kept as the baseline: the same ranking, ten results per page, no filters, and product information on separate sub-pages. PREFLOW uses the same search function, so rankings do not differ between conditions.

S6.3 Tasks and Revision

Both tasks are open: a task card suggests a shopping occasion for participants who have nothing in mind, and participants may look for any product in the shop. Participants decide on their own preferences, such as budget and product properties, and were told that they could change them at any time. For five participants, the researcher said midway through each task: "Imagine your budget has just changed. Your new limit is $\$X$. Please continue with that in mind." A task ends with a purchase or when the participant ends it without buying.

S6.4 Questionnaires

The questionnaires were built for screen readers: each rating item is one labeled group that names both scale ends, there is no time limit, and every page offers an exit.

After each task. Participants rate the task's ease with the Single Ease Question [19] (1, very difficult, to 7, very easy) and workload on the six NASA-TLX dimensions [10]: mental demand, physical demand, temporal demand, performance, effort, and frustration, each from 0 to 100; the workload score is their unweighted mean (raw TLX). They then answer the following items, where *{interface}* is replaced by "shopping on my own" in the Manual condition and "the shopping assistant" in the Agent condition.

Control (1–7, strongly disagree to strongly agree):

(1) I was in control of what happened while I was using {interface}.
(2) I always knew what {interface} was doing.

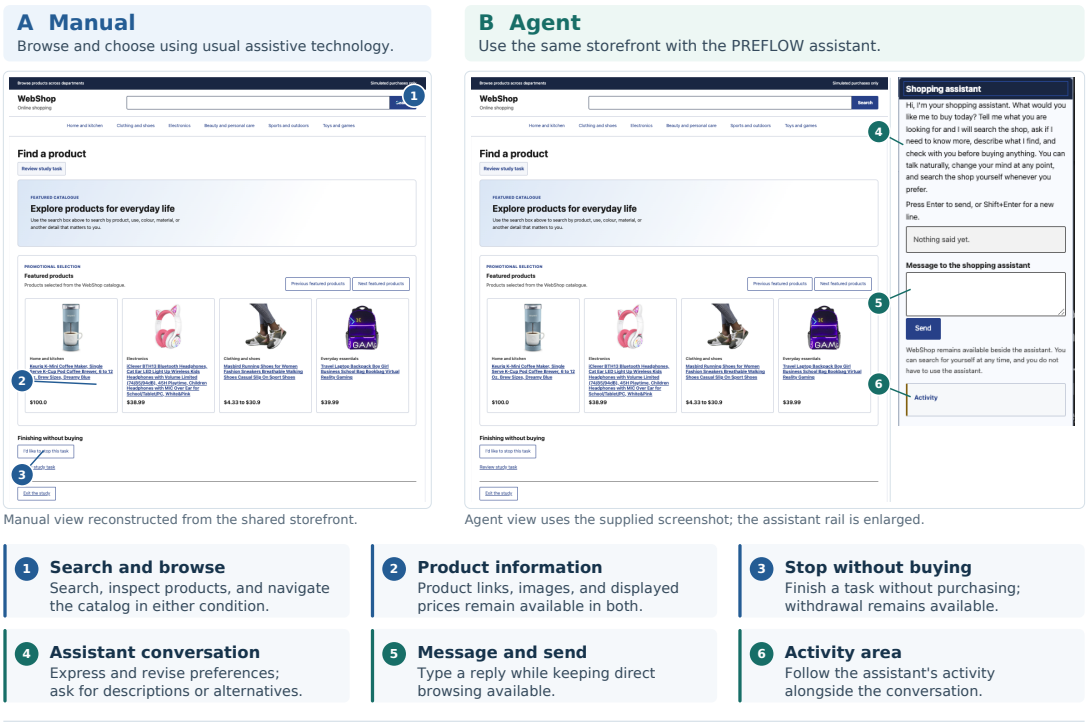


Figure S5: The Manual and Agent conditions use the same shop. The Agent condition adds PREFLOW's conversation panel, shown enlarged for legibility. Both conditions allow direct browsing and task completion without making a purchase. Purchases are simulated.

(3) I could change course whenever I wanted to while using {interface}.

System Usability Scale [2] (1–5), with "awkward" instead of the original "cumbersome" in item 8:

(1) I think that I would like to use {interface} frequently.
(2) I found {interface} unnecessarily complex.
(3) I thought {interface} was easy to use.
(4) I think I would need support from a technical person to use {interface}.
(5) I found the various functions in {interface} were well integrated.
(6) I thought there was too much inconsistency in {interface}.
(7) I would imagine that most people would learn to use {interface} very quickly.
(8) I found {interface} very awkward to use.
(9) I felt very confident using {interface}.
(10) I needed to learn a lot before I could get started with {interface}.

SUS scores range from 0 to 100: each odd-numbered response minus 1 and 5 minus each even-numbered response are summed and multiplied by 2.5 [2].

Trust (1–7, strongly disagree to strongly agree):

(1) I could rely on {interface} to help me find what I wanted.
(2) I would trust {interface} to shop with again.

Product satisfaction. After a purchase, participants rate their satisfaction with the purchased product (1–7).

Final preferences. Participants list the preferences they had at the end of the task and mark, for each, whether they believe the purchased product satisfies it (yes, no, or not sure).

Sessions P4–P8 also include the items in Table S20.

End of session. Participants answer the two items in Table S18 about PREFLOW's messages. They then state whether they prefer shopping with PREFLOW, shopping on their own, or have no preference, explain why, and answer four open questions: "What was the most difficult part of today?", "What worked well for you?", "How sure were you that you bought the right thing?", and "If you could change one thing, what would it be?"

Analysis. We report the mean and standard deviation of each measure per condition and each participant's difference between the conditions, without significance tests. The share of final preferences believed satisfied is computed for each task with a purchase. Interaction logs give the purchases, the questions PREFLOW asked,

Table S18: End-of-session items about PREFLOW's messages, each rated from 1 (low anchor) to 5 (high anchor).

Item	Low / middle / high anchors
How many things did PREFLOW ask you about?	Far too few / About the right number / Far too many
How long were PREFLOW's replies?	Much too short / About the right length / Much too long

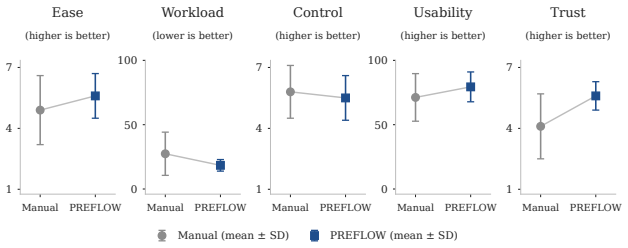


Figure S6: Mean ratings in the Manual condition (shopping on their own) and with PREFLOW; error bars show one standard deviation. Higher is better, except for workload.

Table S19: Each participant's difference, PREFLOW minus Manual, with the mean and median over all eight and the mean without P3, who reported the largest differences. Ease, SUS, and trust: positive is better; workload: negative is better.

	Ease (SEQ)	Workload (TLX)	Usability (SUS)	Trust
P1	0	+9.2	−2.5	+1.5
P2	−3	0.0	−7.5	+2.0
P3	+5	−36.7	+42.5	+4.5
P4	+1	−12.5	+10.0	+1.0
P5	+2	−11.7	+10.0	+1.0
P6	0	−9.2	+5.0	+1.0
P7	+1	−9.2	+2.5	+1.0
P8	0	−1.7	+5.0	0.0
Mean, all eight	+0.8	−9.0	+8.1	+1.5
Median, all eight	+0.5	−9.2	+5.0	+1.0
Mean without P3	+0.1	−5.0	+3.2	+1.1

and the previews. We translated the answers given in Arabic into English. One author grouped the open answers by topic.

S6.5 Results

Table 6 of the main paper and Figure S6 give the means over all eight sessions; Table S19 and Figure S7 give each participant's difference between the conditions. All participants bought a product in both conditions, and every purchase after the budget change stayed within the new budget.

In sessions P4–P8 (Table S20), the largest difference concerns finding out what the products looked like (5.4 with PREFLOW versus 4.0 on their own). Participants also found it easier to go on after changing what they wanted (6.2 versus 5.4), and judged the information equally correct in both conditions (5.6): delegating did not make them doubt what they were told.

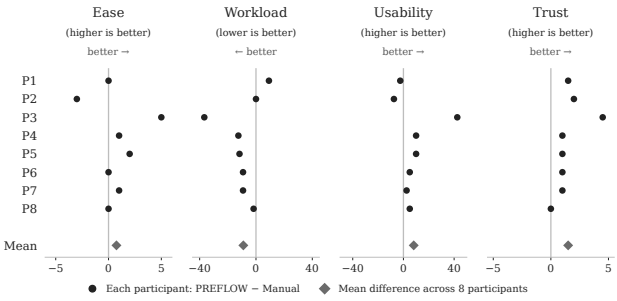


Figure S7: Each participant's difference between the conditions, PREFLOW minus Manual.

Table S20: Items asked in sessions P4–P8 (mean, 1–7; the last item 1–5, 3 is "about right").

Item	Manual	PREFLOW
After I changed what I wanted, it was easy to go on	5.4	6.2
The products offered or found fitted the change	5.8	6.4
I had enough information about the products to decide	4.8	5.4
I could find out what the products looked like	4.0	5.4
The information was correct, as far as I could tell	5.6	5.6
Before the purchase I knew exactly which product, options and price	6.2	6.8
The checks before the purchase were (1 far too few, 5 far too many)	3.0	3.0

S6.6 Who Benefited?

Means over eight participants can hide individual reversals; Table S19 and Figure S7 show four patterns.

Direction. Trust was higher with PREFLOW for seven participants and equal for one. Workload was lower for six, equal for one, and higher for one (P1). Usability was higher for six. Ease was higher for four, equal for three, and lower for one (P2).

Not driven by one participant. P3 reported the largest differences. Without P3, the mean differences still favor PREFLOW for workload (−5.0), usability (+3.2), and trust (+1.1), but not for ease (+0.1). The medians, which P3 alone cannot move, are −9.2 for workload, +5.0 for usability, +1.0 for trust, and +0.5 for ease.

Not confined to chat-assistant users. The three participants who use chat assistants rarely or never (P5, P6, P7; Table S15) all reported lower workload, higher usability, and higher trust with PREFLOW. The two participants whose usability rating dropped (P1, P2) use chat assistants daily and rate their screen-reader proficiency at the maximum. One reading is that highly proficient users gain less from delegation when shopping on their own is already efficient for them; their trust still rose (+1.5, +2.0).

A more uniform experience. The standard deviation across participants fell with PREFLOW for workload (from 16.8 to 4.5), trust (1.6 to 0.7), usability (18.5 to 11.6), and the share of final preferences believed satisfied (0.40 to 0.16; Table 6). Shopping on their own went very differently for different participants; delegating to PREFLOW did not.

With eight participants, these observations are descriptive and do not support statistical claims.

S6.7 Did PREFLOW Ask Too Much, and Did Participants Keep Control?

Questions. PREFLOW asked 9.6 questions per task on average. The study's tasks differ: participants decide what to buy during the task, and the interactive design adds trade-off questions and approval requests (Section 4.5). Participants sent 10.5 messages per task. All eight nevertheless rated the number of questions about right, and in sessions P4–P8 they rated the checks before the purchase about right in both conditions (3.0 on the 1–5 scale; Table S20).

Control. Perceived control was slightly lower with PREFLOW (5.5 versus 5.8). Delegation moves the Web actions to the agent, and one of the three control items asks whether participants always knew what the interface was doing. At the point of decision, however, participants in sessions P4–P8 knew better which product, options, and price they were buying with PREFLOW (6.8 versus 6.2). One reading is that delegation preserved control over decisions but reduced awareness of the agent's ongoing actions, a gap that the activity region (Section S6.2) is meant to address and that future versions should strengthen.

Product appearance. In sessions P4–P8, participants rated how well they could find out what the products looked like higher with PREFLOW (5.4 versus 4.0; Table S20), the need that its photo descriptions address.

What participants said. Participants valued PREFLOW's questions, product descriptions, and comparisons. P2 liked "the questions about more features that I may have not thought about to include in my search" and said, "I could ask more questions if I needed to and it can provide value." For P1, the "description of the product" worked well, and "the correct product is selected." P3 said that PREFLOW "really saves time, helps with choosing once the preferences are set, and helps a lot with comparing products" (translated from Arabic).

S7 Beyond Shopping, and Threats to Validity

In brief. The control flow and the six roles do not depend on shopping; the task schema, the evidence rules, and the checks before the final action do. The main threats to validity are listed below with what we did about them and what remains.

S7.1 What Changes for Another Task Family

Section 7 states that other long-horizon tasks need their own task schema, evidence rules, and checks for the final action, while the control flow and the roles stay the same. Table S21 makes this concrete for the traveler of Section 1, who rebooks a canceled flight while driving to the airport. The mapping is a design sketch; it is neither implemented nor evaluated.

S7.2 Threats to Validity

Table S22 lists the main threats, how we address them, and what remains open.

S7.3 Summary

Across the three settings, the same pattern recurs. Where PREFLOW gains, it gains because the explicit memory keeps what the user wants and what the pages show outside the language model's context: preferences survive long trajectories and a weaker LLM (Sections S4.2 and S4.3), questions target properties that the observed products actually have (Section S4.5), and the report can say what is in doubt before a costly action (Section S5). Where PREFLOW falls short—beliefs that misread a preference, matching rules designed for one shop, a study of eight people—the limitation is explicit and points to the next step.

Table S21: What stays and what changes when PREFLOW moves from shopping to flight rebooking. The rebooking column is a design sketch, not an implemented or evaluated system.

Element	Shopping (evaluated)	Flight rebooking (sketch)
Task schema σ	transactional.obtain: product; attributes; options; other preferences	transactional.obtain: flight; attributes such as arrival time, stops, airline; options such as seat and fare class; other preferences such as price and refundability
Preference sources	Instruction, profile, replies	Instruction ("get me home tonight"), profile (seat preference, loyalty program), replies
Evidence for the Evidence Ledger	Results page, product page, option controls, reviews	Flight results, fare rules, seat map
A typical unresolved preference	How hard a desk is to assemble	Whether checked baggage is transferred
Check before the final action	Product type, price, attributes, options	Arrival time, fare difference, change fees, connection times
Dialogue	Screen-reader announcements; preview of product, options, price, and unmet preferences	Short spoken messages; questions timed to the driver's attention; preview of flight, fare, and unmet preferences
Unchanged	Algorithm 1; the six roles; the statuses supported, contradicted, and unresolved; approval before an action that is costly to reverse; the report of unmet preferences.	

GenAI Usage Disclosure

As for the main paper, we used GPT-6 and Claude Opus 5.5 for organization and language revision. Claude Opus 5.5 also drafted parts of this supplement, including the reader's guide (Table S1), the analyses derived from reported results in Sections S4–S6, and Section S7. The product images in Figure S2 are AI-generated illustrations. The authors reviewed all text and numbers and take full responsibility for them.

References

[1] Andrei Broder. 2002. A taxonomy of web search. *ACM Sigir forum* 36, 2 (2002).

[2] John Brooke. 1996. SUS: A 'Quick and Dirty' Usability Scale. In *Usability Evaluation in Industry*. doi:10.1201/9781498710411-35

[3] Cristiano Castelfranchi and Rino Falcone. 1998. Towards a Theory of Delegation for Agent-Based Systems. *Robotics Auton. Syst.* 24 (1998), 141–157. doi:10.1016/S0921-8890(98)00028-1

[4] Sanxing Chen, Sam Wiseman, and Bhuwan Dhingra. 2024. ChatShop: Interactive Information Seeking with Language Agents. arXiv:2404.09911 doi:10.48550/ARXIV.2404.09911

[5] Xiang Deng, Yu Gu, Boyuan Zheng, et al. 2023. Mind2Web: Towards a Generalist Agent for the Web. In *Proc. of NeurIPS '23*. doi:10.52202/075280-1220

[6] Zeyao Du, Tong Li, and Haibo Zhang. 2026. EComAgentBench: Benchmarking Shopping Agents on Long-Horizon Tasks with Distributed Hidden Intent. arXiv preprint arXiv:2606.17698. arXiv:2606.17698 doi:10.48550/arXiv.2606.17698

[7] Shicheng Fan, Mingdai Yang, Duohao Wang, et al. 2026. Agentic Commerce World: An Auditable and Verifiable Environment for Vibe Commerce. arXiv:2608.02441 [cs.AI] https://arxiv.org/abs/2608.02441

[8] Div Garg, Diego Caples, Andis Draguns, et al. 2025. REAL: Benchmarking Autonomous Agents on Deterministic Simulations of Real Websites. In *Proc. of NeurIPS '25*. doi:10.52202/085713-4536

Table S22: Threats to validity, how we address them, and what remains.

Threat	How we address it	What remains
Search heuristics	Fixed stages and budgets, the same for every task	Not an optimal search policy; all limits are in the released code
Tuning to the benchmarks	No role reads stored goals, rewards, or scripted answers; identical settings for both LLMs; held-out EComAgentBench split (S2.3)	The WebShop Evidence Ledger and verifier use deterministic matching rules; another shop may need other rules
Run-to-run variation	WebShop at temperature 0; paired tests over 500 instructions; intervals for the main estimates (S4.4)	Variance across repeated runs is not estimated; EComAgentBench runs at temperature 1.0
Different LLMs across agents	Same-LLM reruns: five agents with two LLMs on WebShop; the benchmark's own agent with Gemini-3-Flash on EComAgentBench (S4.5)	Published EComAgentBench agents use other LLMs and other evaluator runs
LLM evaluator	Exact target matches bypass the evaluator; in the same-LLM comparison, both agents are judged by the same evaluator	A Gemini evaluator judging Gemini outputs may favor them over the published agents
Development tasks among the 662	100 held-out tasks, never used for development and drawn before any run	The full-benchmark estimate includes the 250 development tasks
Benchmark preferences are not real preferences	BLV study: participants form, revise, and judge their own preferences	WebShop has no user; EComAgentBench's user answers from a script
Manual baseline made harder	Same shop pages in both conditions, so every accessibility fix applies to both; same search and ranking (S6.2)	Manual shopping keeps WebShop's friction: no filters, product information on separate sub-pages
Small, experienced BLV sample	Within-participant, counterbalanced design; per-participant results, medians, and a leave-one-out check (S6.6)	Eight participants, one session each, descriptive results; five of eight use chat assistants daily
Moderated sessions, simulated purchases	Same shop, catalog, and prices in both conditions; approval required; participants could end without buying	Real money may change what participants approve; the moderator's presence may raise ratings in both conditions
Shopping only	Task schemas for other task families (Table S3); design sketch (Table S21)	Not evaluated beyond shopping
Response time	—	18.5 s per reply in the BLV study; likely too slow for a driver

[9] Boyu Gou, Zanming Huang, Yuting Ning, et al. 2025. Mind2Web 2: Evaluating Agentic Search with Agent-as-a-Judge. In *Proc. of NeurIPS '25*. doi:10.52202/085713-5778

[10] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Human Mental Workload*. Vol. 52. doi:10.1016/S0166-4115(08)62386-9

[11] Bernard J. Jansen, Danielle L. Booth, and Amanda Spink. 2008. Determining the informational, navigational, and transactional intent of Web queries. *Inf. Process. Manag.* 44, 3 (2008). doi:10.1016/J.IPM.2007.07.015

[12] Sunghwan Kim, Ryang Heo, Yongsik Seo, et al. 2026. AgenticShop: Benchmarking Agentic Product Curation for Personalized Web Shopping. In *Proc. of WWW '26*. doi:10.1145/3774904.3792724

[13] Serin Kim, Sangam Lee, and Dongha Lee. 2026. Persona2Web: Benchmarking Personalized Web Agents for Contextual Reasoning with User History. arXiv preprint. arXiv:2602.17003 https://arxiv.org/abs/2602.17003v1

[14] Xing Han Lù, Zdenek Kasner, and Siva Reddy. 2024. WebLINX: Real-World Website Navigation with Multi-Turn Dialogue. In *Proc. of ICML '24*. https://proceedings.mlr.press/v235/lu24e.html

[15] Kaixin Ma, Hongming Zhang, Hongwei Wang, et al. 2024. LASER: LLM Agent with State-Space Exploration for Web Navigation. arXiv preprint. Version 2, 21 February 2024. arXiv:2309.08172 doi:10.48550/arXiv.2309.08172

[16] Ralph Peeters, Aaron Steiner, Luca Schwarz, et al. 2026. WebMall – A Multi-Shop Benchmark for Evaluating Web Agents. In *Proc. of SIGIR '26*. doi:10.1145/3805712.3808592

[17] Xavier Puig, Kevin Ra, Marko Boben, et al. 2018. VirtualHome: Simulating Household Activities via Programs. In *Proc. of CVPR '18*. doi:10.1109/CVPR.2018.00886

[18] Cheng Qian, Zuxin Liu, Akshara Prabhakar, et al. 2025. UserBench: An Interactive Gym Environment for User-Centric Agents. arXiv preprint arXiv:2507.22034. arXiv:2507.22034 [cs.AI] doi:10.48550/arXiv.2507.22034

[19] Jeff Sauro and Joseph S. Dumas. 2009. Comparison of Three One-Question, Post-Task Usability Questionnaires. In *Proc. of CHI '09*.

[20] Tianlin Shi, Andrej Karpathy, Linxi Fan, et al. 2017. World of Bits: An Open-Domain Platform for Web-Based Agents. In *Proc. of ICML '17*. http://proceedings.mlr.press/v70/shi17a.html

[21] Daniel Toyama, Philippe Hamel, Anita Gergely, et al. 2021. AndroidEnv: A Reinforcement Learning Platform for Android. arXiv preprint arXiv:2105.13231. arXiv:2105.13231 doi:10.48550/arXiv.2105.13231

[22] Shunyu Yao, Howard Chen, John Yang, et al. 2022. Webshop: Towards scalable real-world web interaction with grounded language agents. In *Proc. of NeurIPS '22*.

[23] Jingheng Ye, Yong Jiang, Xiaobin Wang, et al. 2025. ProductAgent: Benchmarking Conversational Product Search Agent with Asking Clarification Questions. In *Proc. of EMNLP '25 (Industry Track)*. doi:10.18653/v1/2025.emnlp-industry.25

[24] Suyu Ye, Haojun Shi, Darren Shih, et al. 2026. RealWebAssist: A Benchmark for Long-Horizon Web Assistance with Real-World Users. In *Proc. of AAAI '26*. doi:10.1609/AAAI.V40I40.40742

[25] Shuyan Zhou, Frank F. Xu, Hao Zhu, et al. 2024. WebArena: A Realistic Web Environment for Building Autonomous Agents. In *Proc. of ICLR '24*. https://openreview.net/forum?id=oKn9c6ytLx